

SCHOOL RESULT PORTAL

A Web-Based Academic Result Management System

Capstone Project Proposal

Prepared By: DevCore 7-Innovators Team

Table of Contents

- 1. Introduction**
- 2. Problem Statement**
- 3. Aim of the Project**
- 4. Objectives**
- 5. Scope of the Project**
- 6. System Users**
- 7. Functional Requirements**
- 8. Non-Functional Requirements**
- 9. System Features**
- 10. Technologies to be Used**
- 11. System Architecture**
- 12. Database Design**
- 13. Methodology**
- 14. Project Timeline**
- 15. Expected Deliverables**
- 16. Benefits of the System**
- 17. Limitations**
- 18. Future Improvements**
- 19. Conclusion**



1. Introduction

Educational institutions generate and manage a large amount of students' academic records every session. Traditionally, many schools rely on spreadsheets, paper-based records, or standalone desktop applications to compute and manage students' results. These methods are often inefficient, error-prone, and difficult to maintain.

The School Result Portal is a web-based application designed to automate the process of recording, computing, storing, and publishing students' academic results. The system provides secure access for administrators, teachers, students, and parents while ensuring data accuracy, confidentiality, and availability.

2. Problem Statement

Many schools still depend on manual methods for processing students' academic results. These methods present several challenges, including:

- Time-consuming result computation.
- Human errors during grading and calculations.
- Poor storage and retrieval of student records.
- Delayed publication of examination results.
- Limited accessibility for students and parents.
- Weak security of academic records.
- Difficulty in generating report cards.
- Poor monitoring of students' academic performance.

These challenges create inefficiencies that negatively affect school administration and decision-making.

3. Aim of the Project

The aim of this project is to design and develop a secure web-based application that

automates the management, computation, storage, and publication of students' academic results.

4. Objectives

The objectives of this project are to:

- Develop a secure authentication system.
- Create separate dashboards for administrators, teachers, students, and parents.
- Manage students, teachers, classes, subjects, academic sessions, and terms.
- Enable teachers to upload Continuous Assessment (CA) and examination scores.
- Automatically calculate total scores, grades, averages, GPA (if applicable), and class positions.
- Generate printable report cards.
- Allow students to view and download their results online.
- Provide analytical reports to administrators.
- Maintain secure and reliable academic records.

5. Scope of the Project

The system will cover:

Administrator Module

- User Management
- Student Registration
- Teacher Management
- Subject Management
- Class Management



- Academic Session Management
- Result Approval
- Dashboard Statistics
- Reports

Teacher Module

- Login
- Upload CA Scores
- Upload Examination Scores
- Edit Submitted Scores
- Submit Results
- View Assigned Classes

Student Module

- Secure Login
- View Results
- Download Report Cards
- View Academic History

Parent Module (Optional)

- View Child's Results
- Monitor Academic Progress

6. System Users

The system consists of four major categories of users:



Administrator

Responsible for managing all school records, approving results, and maintaining the system.

Teacher

Responsible for entering students' scores and submitting results.

Student

Responsible for viewing and downloading personal academic results.

Parent

Responsible for monitoring their child's academic performance.

7. Functional Requirements

The system shall:

- Authenticate users securely.
- Register students and teachers.
- Create classes and subjects.
- Assign subjects to teachers.
- Record Continuous Assessment scores.
- Record examination scores.
- Compute total scores automatically.
- Calculate grades automatically.
- Compute averages.



- Compute class positions.
- Generate report cards.
- Allow result downloads.
- Produce analytical reports.
- Maintain historical records.

8. Non-Functional Requirements

The system should:

- Be secure.
- Be reliable.
- Be scalable.
- Be responsive.
- Be user-friendly.
- Provide fast response times.
- Ensure high availability.
- Protect confidential information.
- Support multiple devices.

9. System Features

- Secure Authentication
- Role-Based Access Control
- Student Information Management
- Teacher Management
- Subject Management



- Academic Session Management
- Continuous Assessment Entry
- Examination Score Entry
- Automatic Grade Computation
- Automatic Position Calculation
- Report Card Generation (PDF)
- Dashboard Analytics
- Search and Filtering
- Responsive User Interface

10. Technologies to be Used

Frontend

- React.js
- JavaScript
- HTML5
- CSS3

Backend

- Supabase

Database

- PostgreSQL (Supabase)

Authentication



- Supabase Authentication

Deployment

- Vercel or Netlify (Frontend)
- Supabase Cloud (Backend)

Version Control

- Git
- GitHub

11. System Architecture

The proposed system follows a three-tier architecture.

Presentation Layer:

- React.js User Interface

Application Layer:

- Supabase Backend Services
- Authentication
- Business Logic

Data Layer:

- PostgreSQL Database



Users interact with the frontend, which communicates securely with Supabase services. Supabase processes requests and stores or retrieves information from the PostgreSQL database.

12. Database Design

Major database tables include:

- Users
- Students
- Teachers
- Classes
- Subjects
- Sessions
- Terms
- Results
- Scores
- Report Cards

Relationships:

- One class has many students.
- One teacher teaches many subjects.
- One student has many results.
- One session contains many terms.
- One subject belongs to multiple classes.



13. Methodology

The project follows the Agile Software Development Methodology.

Development phases include:

1. Requirements Gathering
2. System Analysis
3. Database Design
4. UI/UX Design
5. Frontend Development
6. Backend Development
7. Database Integration
8. Testing
9. Deployment
10. Documentation

Agile enables continuous testing, feedback, and improvements throughout development.

14. Project Timeline

Phase| Duration

Requirements Gathering| 1 Week

System Analysis| 1 Week

Database Design| 1 Week

UI/UX Design| 2 Weeks



Frontend Development| 3 Weeks

Backend Development| 3 Weeks

Integration| 1 Week

Testing| 2 Weeks

Deployment| 1 Week

Documentation| 1 Week

Estimated Total Duration: 16 Weeks

15. Expected Deliverables

The project will deliver:

- Fully Functional School Result Portal
- Responsive Web Application
- PostgreSQL Database
- RESTful Backend Services
- Technical Documentation
- User Manual
- GitHub Repository
- Deployment Guide

16. Benefits of the System

The system will:

- Reduce errors in result computation.



- Improve processing speed.
- Enhance data security.
- Simplify record management.
- Provide easy online access to results.
- Reduce paperwork.
- Improve academic decision-making.
- Increase transparency and accountability.

17. Limitations

Some limitations include:

- Internet connection is required.
- Initial user training may be necessary.
- Data migration from existing manual systems may require additional effort.
- System performance depends on server availability.

18. Future Improvements

Future versions may include:

- SMS Result Notification
- Email Notifications
- Parent Portal
- Mobile Application
- Attendance Management



- School Fee Management
- Student Transcript Generation
- AI-Based Performance Prediction
- Data Visualization Dashboard
- Multi-School Support

19. Conclusion

The School Result Portal is designed to modernize academic result management by replacing manual processes with a secure, reliable, and efficient web-based system. The application will significantly reduce errors, improve result processing speed, enhance data security, and provide timely access to academic records for administrators, teachers, students, and parents.

By leveraging modern technologies such as React.js, Supabase, and PostgreSQL, the proposed system offers a scalable and user-friendly solution capable of meeting the academic management needs of educational institutions.

Prepared By : DevCore 7-Innovators Team

Capstone Project

Project Title: School Result Portal – A Web-Based Academic Result Management System

